

7(a)	output voltage / input voltage	M1
	input (voltage) is difference between (inverting and non-inverting) inputs	A1
7(b)	- reduces the gain - greater bandwidth - more stable <i>Any two points, 1 mark each</i>	B2
7(c)(i)	inverting amplifier	B1
7(c)(ii)	X marked anywhere between right-hand edge of $480\ \Omega$ resistor, left-hand edge of $1.2\ \text{k}\Omega$ resistor and the inverting input	B1
7(c)(iii)	gain = $(-)R_f / R_i$	C1
	= $(-)1200 / 480$	A1
	= -2.5	
7(c)(iv)	$V_{\text{IN}} = 6.5 / (-2.5)$ $= -2.6\ \text{V}$	A1
7(c)(v)	$(-2.5) \times (-5.4) = +13.5\ \text{V}$, and so output saturates $V_{\text{OUT}} = (+)8.0\ \text{V}$	A1